

Trabalho 1 - Compiladores

Regras de produção escaladas

4. var-declaration $\rightarrow$ type-specifier ID ; | type-specifier ID ENUM

18. expression $\rightarrow$ var = expression | simple-expression

21. addop $\rightarrow$ + | -

25. multop $\rightarrow$ * | /

27. ull $\rightarrow$ ID (args)

NUM e ID também

Terminais de cada regra:

4. ID, [,], NUM

25. *, /

18. =

27. ID, (,)

23. +, -

Expressões Regulares

ID = letter letter*

NUM = digit digit*

ID [NUM] = ID [NUM]

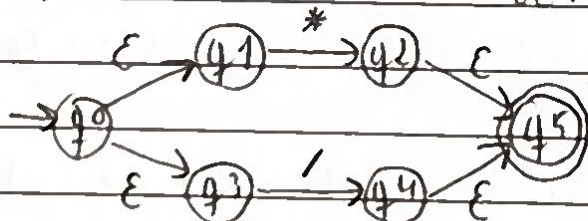
Regras de produção 25:

ER: * | /

tokens: *, /

NFA:

(Thompson)



DFA por ε-fechas:

$\epsilon(q_0) = \{q_0, q_1, q_3\}$, $\epsilon(q_1) = \{q_1\}$, $\epsilon(q_2) = \{q_2, q_5\}$, $\epsilon(q_3) = \{q_3\}$

spiral $\epsilon(q_4) = \{q_4, q_5\}$, $\epsilon(q_5) = \{q_5\}$

$$0 = \varepsilon(q_0) = \{q_0, q_1, q_3\}$$

$$0 \text{ lê } "*" : \varepsilon(q_2) = 1 \text{ (nova estaca)} = \{q_2, q_5\}$$

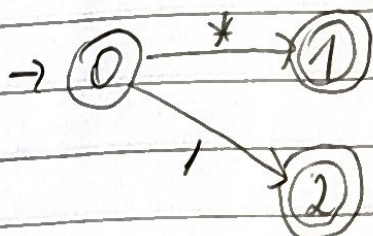
$$0 \text{ lê } "/" : \varepsilon(q_4) = 2 \text{ (nova estaca)} = \{q_4, q_5\}$$

$$1 \text{ lê } "*" : \varepsilon(\emptyset)$$

$$1 \text{ lê } "/" : \varepsilon(\emptyset)$$

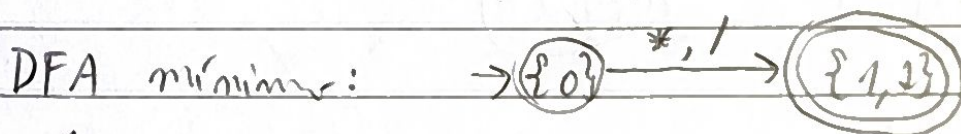
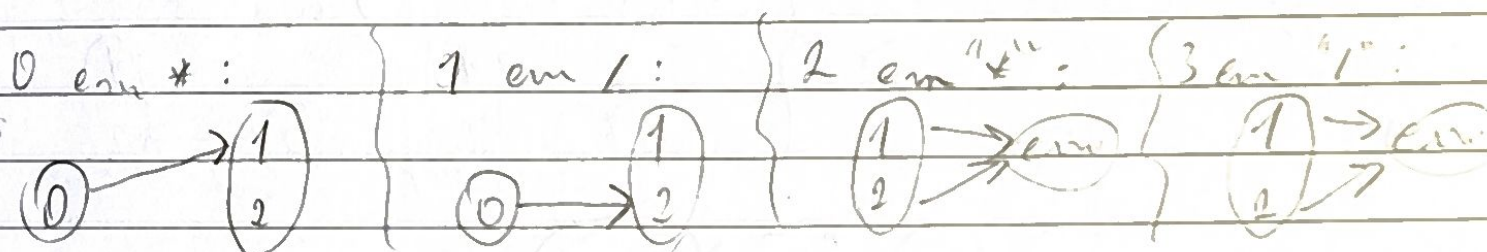
$$2 \text{ lê } "*" : \varepsilon(\emptyset)$$

$$2 \text{ lê } "/" : \varepsilon(q)$$



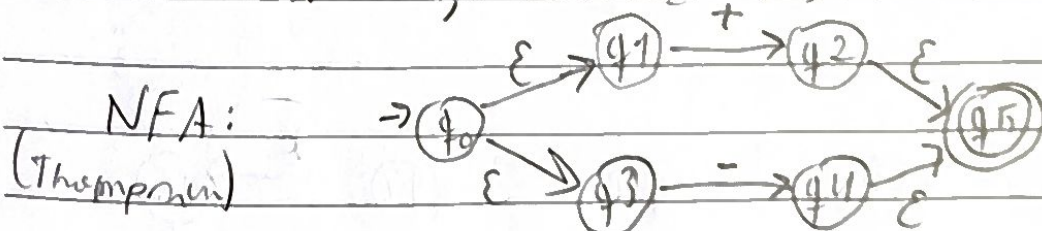
DFA mínima por Hopcroft

	Part	0	Split
0	<u>{0}</u> {1,2}	*	—
1	{0} <u>{1,2}</u>	/	—
2	{0} <u>{1,2}</u>	*	—
3	{0} <u>{1,2}</u>	/	—



Regra de produção 23:

ER: + | - ; Tokens: +, -



DFA por E-frecho: $\varepsilon(q_0) = \{q_0, q_1, q_3\}$, $\varepsilon(q_1) = \{q_1\}$, $\varepsilon(q_2) = \{q_2, q_5\}$, $\varepsilon(q_3) = \{q_3\}$, $\varepsilon(q_4) = \{q_4, q_5\}$, $\varepsilon(q_5) = \{q_5\}$

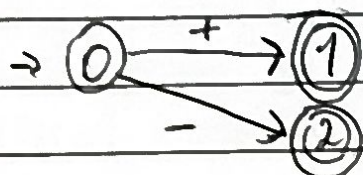
$$O = \varepsilon(q_0) = \{q_0, q_1, q_2\}$$

$$O \text{ lê "+" : } \varepsilon(q_2) = \{q_2, q_5\} = 1$$

$$O \text{ lê "-" : } \varepsilon(q_4) = \{q_4, q_5\} = 2$$

$$1 \text{ lê "+" : } \varepsilon(\emptyset), \quad 1 \text{ lê "-" : } \varepsilon(\emptyset)$$

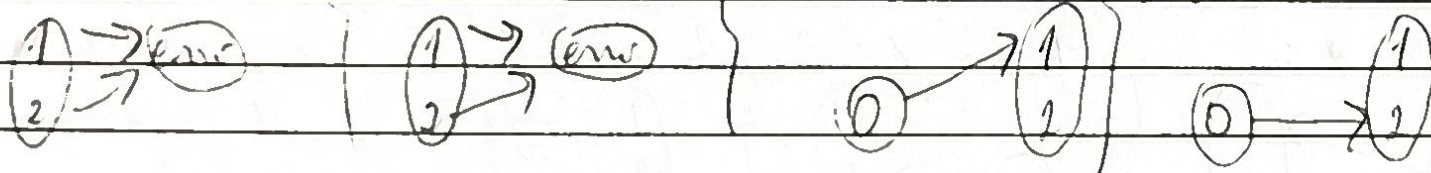
$$2 \text{ lê "+" : } \varepsilon(\emptyset), \quad 2 \text{ lê "-" : } \varepsilon(\emptyset)$$



DFA mínimo por Hopcroft

	Part	0	Split
0	{0} {1,2}	+	—
1	{0} {1,2}	-	—
2	{0} {1,2}	+	—
3	{0} {1,2}	-	—

0 em "+": 1 em "-": 2 em "+": 3 em "-":

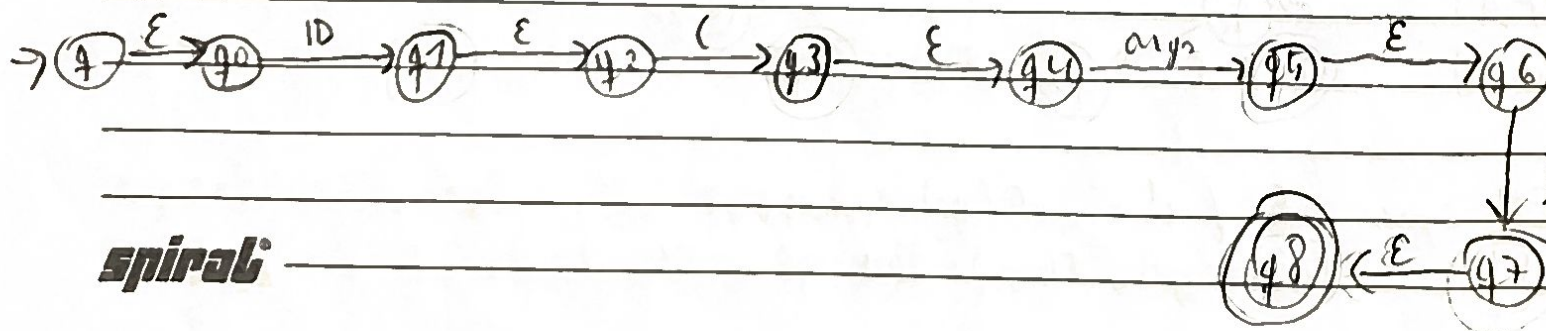


DFA mínimo: $\rightarrow \{0\} \xrightarrow{+, -} \{1, 2\}$

Regra de Produção 27

ER: $ID(mys)$

NFA por Thompson:



DFA per ϵ -fächer

$$\epsilon(q) = \{q, q_0\}$$

$$\epsilon(q_0) = \{q_0\}, \epsilon(q_1) = \{q_1, q_2\}, \epsilon(q_2) = \{q_2\}, \epsilon(q_3) = \{q_3, q_4\}, \epsilon(q_4) = \{q_4\}$$

$$\epsilon(q_5) = \{q_5, q_6\}, \epsilon(q_6) = \{q_6\}, \epsilon(q_7) = \{q_7, q_8\}, \epsilon(q_8) = \{q_8\}$$

$$0 = \epsilon(q) = \{q, q_0\}$$

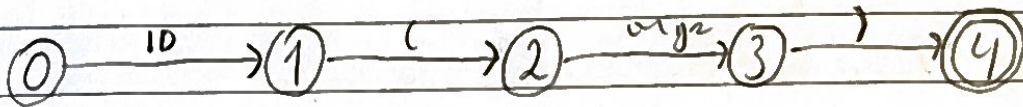
$$0 \text{ ist "ID": } \epsilon(q_1) = \{q_1, q_2\} = 1; 0 \text{ ist } \neg \text{ID: } \epsilon(\emptyset)$$

$$1 \text{ ist "(" : } \epsilon(q_3) = \{q_3, q_4\} = 2; 1 \text{ ist } \neg \text{"(" : } \epsilon(\emptyset)$$

$$2 \text{ ist "arg": } \epsilon(q_5) = \{q_5, q_6\} = 3; 2 \text{ ist } \neg \text{arg: } \epsilon(\emptyset)$$

$$3 \text{ ist ")" : } \epsilon(q_7) = \{q_7, q_8\} = 4; 3 \text{ ist } \neg \text{") : } \epsilon(\emptyset)$$

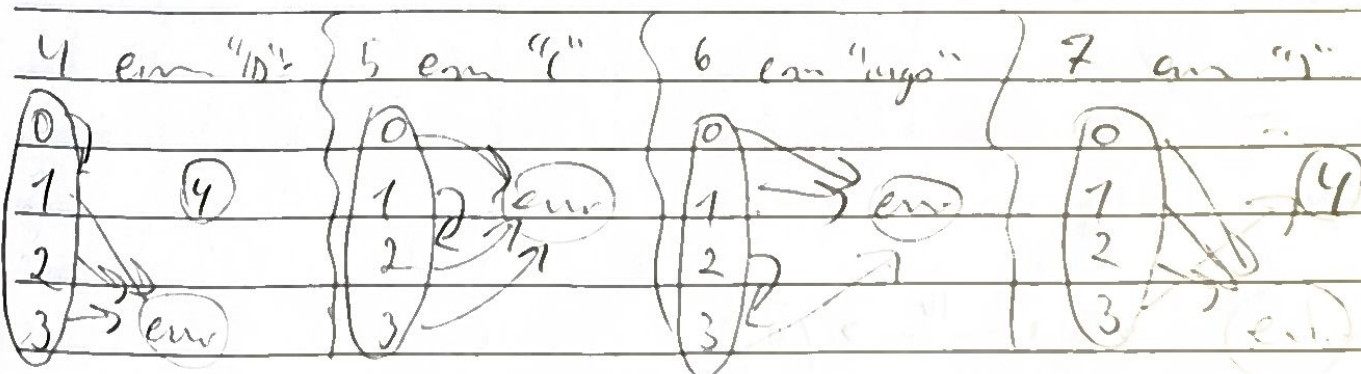
$$4 \text{ ist ID | (| arg |) : } \epsilon(\emptyset)$$



DFA minimierung per Hopcroft:

	Part	Q	Split	
0	$\{0, 1, 2, 3\} \underline{\{4\}}$	ID	—	
1	$\{0, 1, 2, 3\} \underline{\{4\}}$	(	—	
2	$\{0, 1, 2, 3\} \underline{\{4\}}$	arg	—	
3	$\{0, 1, 2, 3\} \underline{\{4\}}$	)	—	
4	$\{0, 1, 2, 3\} \underline{\{4\}}$	ID	—	
5	$\{0, 1, 2, 3\} \underline{\{4\}}$	(	—	
6	$\{0, 1, 2, 3\} \underline{\{4\}}$	arg	—	
7	$\{0, 1, 2, 3\} \underline{\{4\}}$	)	—	

$$0 \text{ em "ID": } \{4 \rightarrow \text{err}\} \quad 1 \text{ em "(" : } \{4 \rightarrow \text{err}\} \quad 2 \text{ em "arg": } \{4 \rightarrow \text{err}\} \quad 3 \text{ em ")" : } \{4 \rightarrow \text{err}\}$$

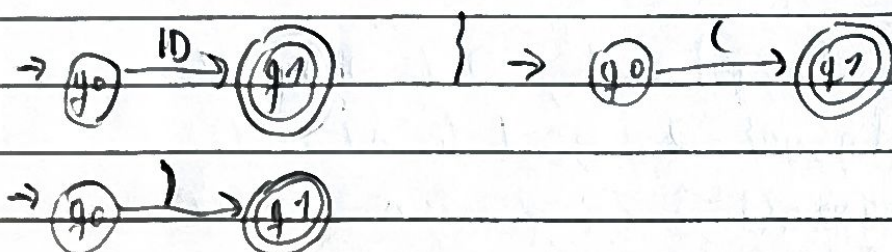


1 / 1

Regra de Produção 27

ER: 1. ID 2. (; tokens: ID, (,)
3.)

NFA por Thompson

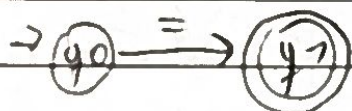


Todas são DFAs mínimas

Regra de Produção 18

ER: = tokens: =

NFA por Thompson

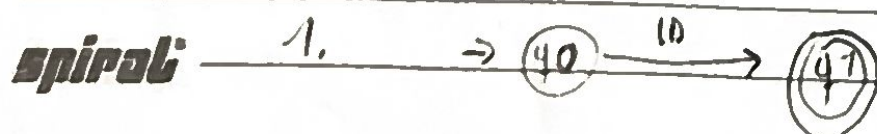


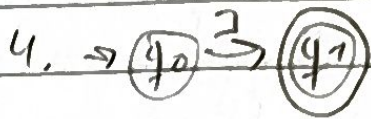
Se é uma DFA mínima

Regra de Produção 4

ER: 1. ID 2. [3. NUM 4.] ; tokens: ID, [, NUM,]

NFA por Thompson



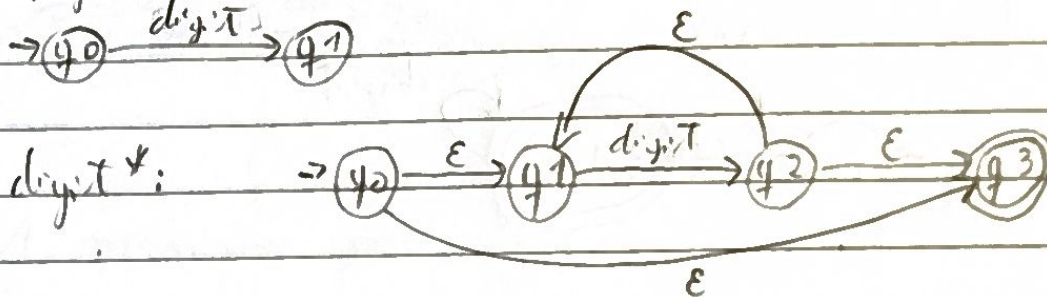
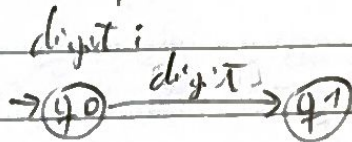


total für nur DFA minimieren

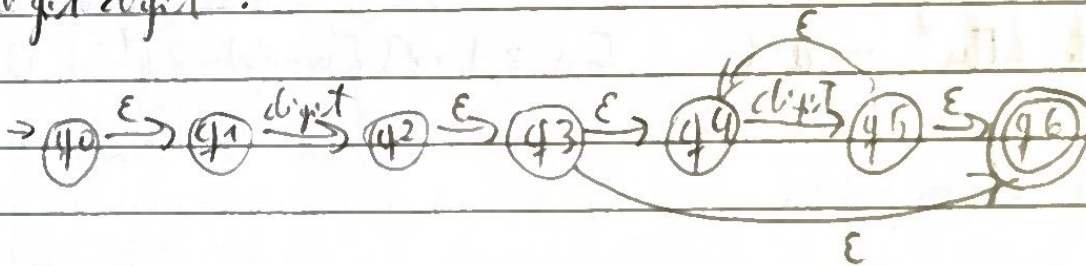
NUM:

ER: $[0-9][0-9]^*$ or $digit\ digit^*$

NFA per Thompson:



digit digit*:



DFA per ϵ -closure

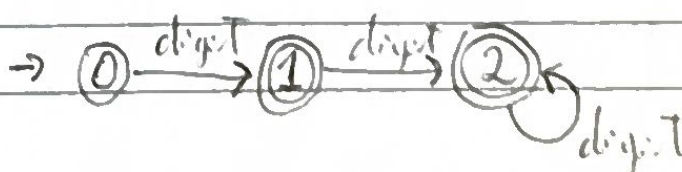
$\epsilon(q_0) = \{q_0, q_1\}$, $\epsilon(q_1) = \{q_1\}$, $\epsilon(q_2) = \{q_2, q_3, q_4, q_6\}$, $\epsilon(q_3) = \{q_3, q_4, q_6\}$, $\epsilon(q_4) = \{q_4, q_6\}$, $\epsilon(q_5) = \{q_5, q_4, q_6\}$, $\epsilon(q_6) = \{q_6\}$

0 = $\epsilon(q_0) = \{q_0, q_1\}$

0 em digit: $\epsilon(q_2) = \{q_2, q_3, q_4, q_6\} = 1$ (more states)

1 em digit: $\epsilon(q_5) = \{q_5, q_4, q_6\} = 2$ (more states)

2 em digit: $\epsilon(q_5) = 2$ (states existents)

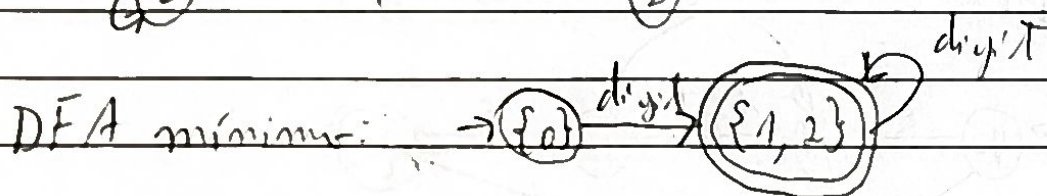
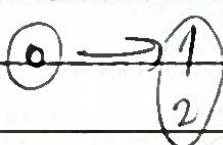
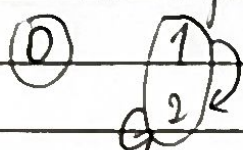


/ /

DFA minimiere per Merschnitt

	Part	0	1
0	$\{0\} \cdot \{1,2\}$	digit	—
1	$\{0\} \cdot \{1,2\}$	digit	—

0 ein "digit": { 1 ein digit



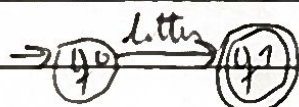
— \

ID:

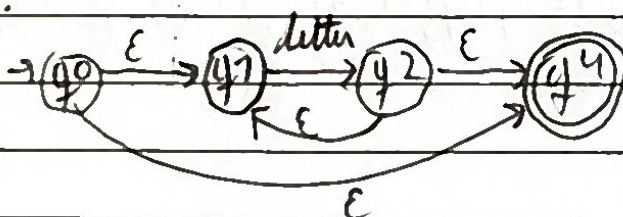
ER: letter letter* all [a-zA-Z][a-zA-Z]*

NFA per Thompson:

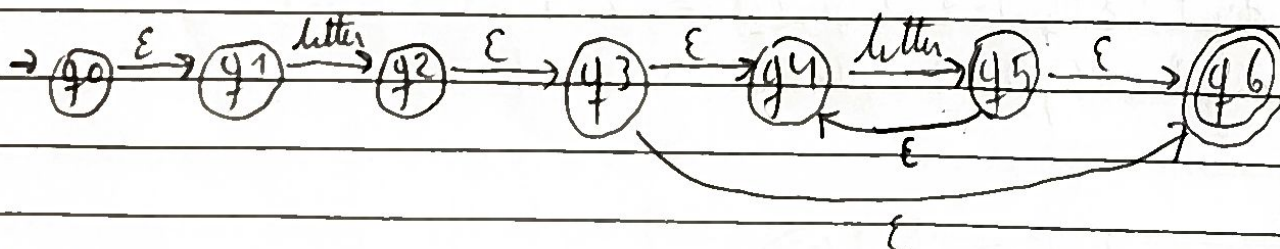
letter:



letter*:



letter letter*:



DFA per E-fuchs

$$\begin{aligned} \varepsilon(q_0) &= \{q_0, q_1\}, \varepsilon(q_1) = \{q_1\}, \varepsilon(q_2) = \{q_2, q_3, q_4, q_6\}, \varepsilon(q_3) = \{q_3, q_4, q_6\} \\ \varepsilon(q_4) &= \{q_4\}, \varepsilon(q_5) = \{q_5, q_4, q_6\}, \varepsilon(q_6) = \{q_6\} \end{aligned}$$

$$0 = \varepsilon(q_0) = \{q_0, q_1\}$$

$$0 \text{ em letter: } \varepsilon(q_2) = \{q_2, q_3, q_4, q_6\} = 1 \text{ (never reached)}$$

$$1 \text{ em letter: } \varepsilon(q_5) = \{q_5, q_4, q_6\} = 2 \text{ (never reached)}$$

$$2 \text{ em letter: } \varepsilon(q_5) = 2 \text{ (reached existing)}$$



DFA minimi per Mapcraft

	Part	0	Split
0	$\{0\} - \{1,2\}$	letter	—
1	$\{0\} - \{1,2\}$	letter	—

